Lab 4

Objective: Practice **Streams, Basic File I/O, Character I/O**.

In this lab, you are going to practice I/O streams and character processing.

Task 1: practice simple file input/output by using the 6.1 sample code, you should create an input file called "infile.dat", which looks like this:

```
infile - Notepad
File  Edit  Format  View  Help
34  56  89
200   40
12
```

Finish the following code and run the program, check what's in the output file called "outfile.dat".

```cpp
//Reads three numbers from the file infile.dat, sums the numbers,
//and writes the sum to the file outfile.dat.
#include <fstream>
#include <iostream>
#include <cstdlib>

int main()
{
    using namespace std;
    ifstream inStream;
    ofstream outStream;

    inStream.open("infile.dat");
    if (inStream.fail())
    {
        cout << "Input file opening failed.\n";
        exit(1);
    }

    outStream.open("outfile.dat");
    if (outStream.fail())
    {
        cout << "Output file opening failed.\n";
        exit(1);
    }

    int first, second, third;
    inStream >> first >> second >> third;
    outStream << "The sum of the first 3\n"
        << "numbers in infile.dat\n"
        << "is " << (first + second + third)
        << endl;

    inStream.close();
    outStream.close();

    return 0;
}
```

<u>**Task 2:**</u> practice character I/O, you should create an input file called "in.dat", which looks like this:

```
in - Notepad
File  Edit  Format  View  Help
12,45,67,300,60
```

Finish the following code and run the program, check what's in the output file called "out.dat".

```cpp
#include <iostream>
#include <fstream>
#include <cctype>  // Include cctype for character classification
#include <string>
using namespace std;
int main() {

    ifstream inStream;
    ofstream outStream;

    inStream.open("in.dat");
    if (inStream.fail())
    {
        cout << "Input file opening failed.\n";
        exit(1);
    }

    outStream.open("out.dat");
    if (outStream.fail())
    {
        cout << "Output file opening failed.\n";
        exit(1);
    }

    string number = "";  // To store each number as we extract them
    int sum = 0;
    char ch;

    // Read character by character from the file
    while (inStream.get(ch)) {
        if (isdigit(ch)) {  // If the character is a digit, add it to the number string
            number += ch;
        }
        else if (ch == ',' || inStream.eof()) {  // If a comma is encountered, process the number
            if (!number.empty()) {
                sum += stoi(number);  // Convert the string to an integer and add to sum
                outStream << number<<"\n";
                number = "";  // Reset the number string for the next number
            }
        }
    }

    // Handle the last number if the file does not end with a comma
    if (!number.empty()) {
        outStream << number << "\n";
        sum += stoi(number);
    }
    outStream << "The sum of the numbers are: " << sum;


    // Close the files

    inStream.close();
    outStream.close();

    return 0;
}
```

Finish the following code and run the program, check what's in the output file called "out.dat".

<u>Submission:</u>

You need to upload: <u>the source code (.cpp files)</u>, <u>your input text files</u> and <u>the output files</u> on Moodle before 4pm Wednesday.